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EDUCATION

BS Mechanical Engineering

Expected May 2029

Johns Hopkins University

GPA: 3.95 / 4.0, Dean's List

Courses: Python; Linear Algebra; Differential Equations; Design/CAD; MATLAB; Thermodynamics; Statics/Mechanics

EXPERIENCE

Weihs Lab Researcher - Johns Hopkins Extreme Materials Institute

April 2026 - Present

- Operated furnace to heat treat Mg-Al rods up to 450 C; executed 0.1-mm tolerance **CNC Wire EDM** to cut 1-2 mm thick samples, optimizing **g-code** and clamping methods to accelerate project completion by **25%**
- Manufactured parts for 3-point bending machine, including threaded resin-printed load grip and fluid funnel
- Adopted PhD student's job in applying precise **lacquer** to all contact points of **micrometer**-measured samples to prevent unwanted crack propagation during cyclic loading and ensure evenly-spaced grips
- Led 4 corrosion-only experiments to corroborate corrosion rate acquired from 3-point bending, designing 2-minute-assembly clamping apparatus in **SolidWorks** to suspend 12-15 5x5x2 mm Mg samples in corrosive salt solution for three days such that the mass loss and pH for each sample stay in specific ranges

Product & Data Lead at Kamoky (Internship at Social App for Ethnic Products)

May 2026 - August 2026

- Built live, RAM-optimized **Python-Flask** dashboard CEO views for free, classifying past 30 days of posts into user and business categories while computing retention metrics aggregated by city, month, and topic
- Performed detailed **SQL** queries to discover in-app gaps between diaspora users' needs and advertised products, culminating in first-kind report shown to leaders of U.S.-African government consultancies
- Aided ethnic businesses in learning about their posts' reach and viewer info (i.e. topics viewed most, location, retention by location, activity by month); developed under **MATLAB** GUI and Python backend
- Devised accurate, instant logging of clicks on app store links promoted by media influencers, including conversion to app downloads and log-in, by leveraging **Mixpanel-MongoDB** device metadata matching

Researcher in Johns Hopkins Multi-Hazard Resilient Structures Lab

January 2026 - Present

- Leading novel prediction of temperature in cold-form steel structures under fire by simplified physics model
- Developed multiple **2D FEA** thermal models and parametric analyses of gypsum-protected, cold-form steel wall stud assemblies exposed to experimental fires monitored by Dr. Gernay (PI) and other fire researchers
- Presented graduate research group with validation of measured experimental temperatures using FE results

Design Build Fly Club (AIAA Competition Team) - Johns Hopkins University

September 2025 - Present

- Conducted carbon-fiber torsion analysis (**ANSYS**) to minimize stress and cost of fuselage for chief engineer
- Led manufacturing process for banner frame and deployment; designed linear actuator mechanism in **SolidWorks** to unfurl and drop the banner; rapidly prototyped to prevent release malfunction during flight
- Successful banner mission yields most score, placing us **17th out of 175 global teams** (best in club history)

LEADERSHIP

President of Futurism at JHU Podcast

January 2026 - Present

- 100,000+ listens, 30+ countries, episodes with leaders in neurology, climate, cybersecurity, biodiversity, AI

Business Development Associate for Lumiere Education

October 2025 - April 2026

- Communicated with 100+ students, clubs, faculty, and career center to promote remote work opportunities

SKILLS AND CERTIFICATES

CAD: Onshape, SolidWorks, Fusion, NX; **FEA:** ANSYS; **MFG:** EDM, 3D/Resin Printing, Laser Cutting, Drill, Band Saw
Map The System (**Systems Thinking**) Program Certificate - Skoll Centre for Social Entrepreneurship, University of Oxford